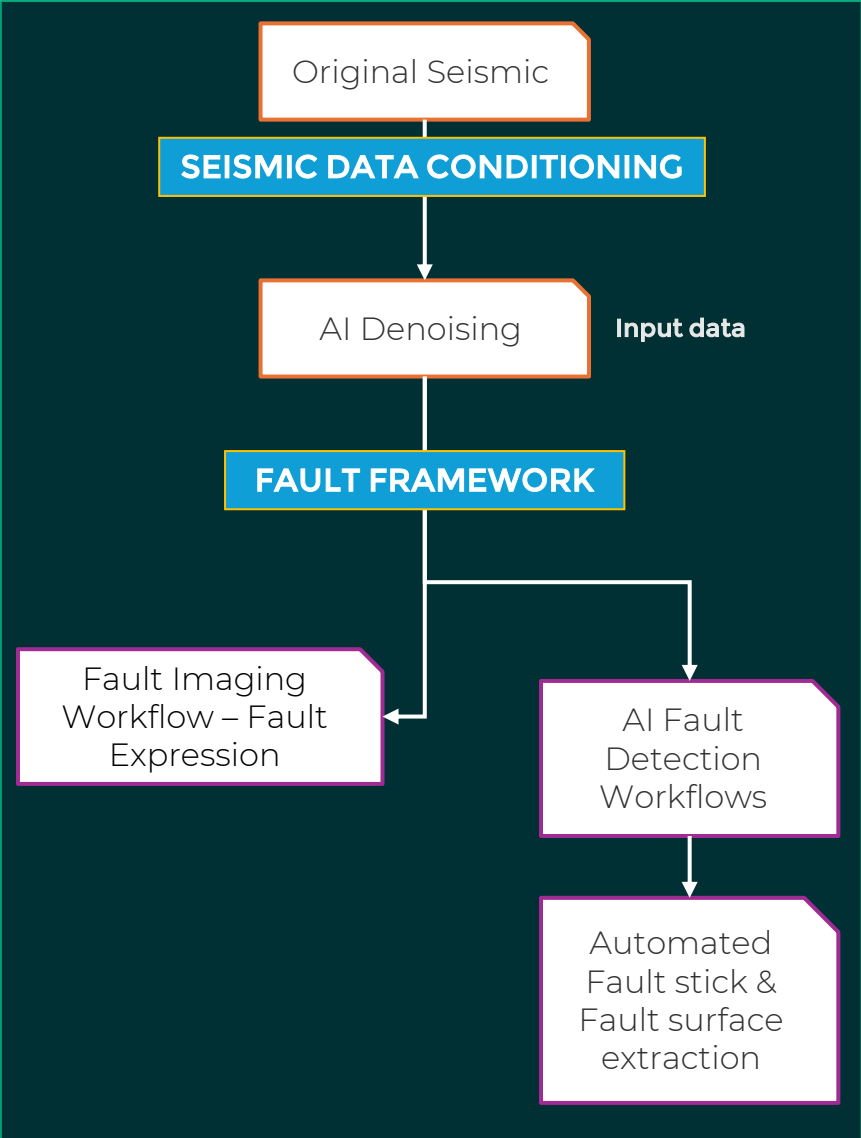
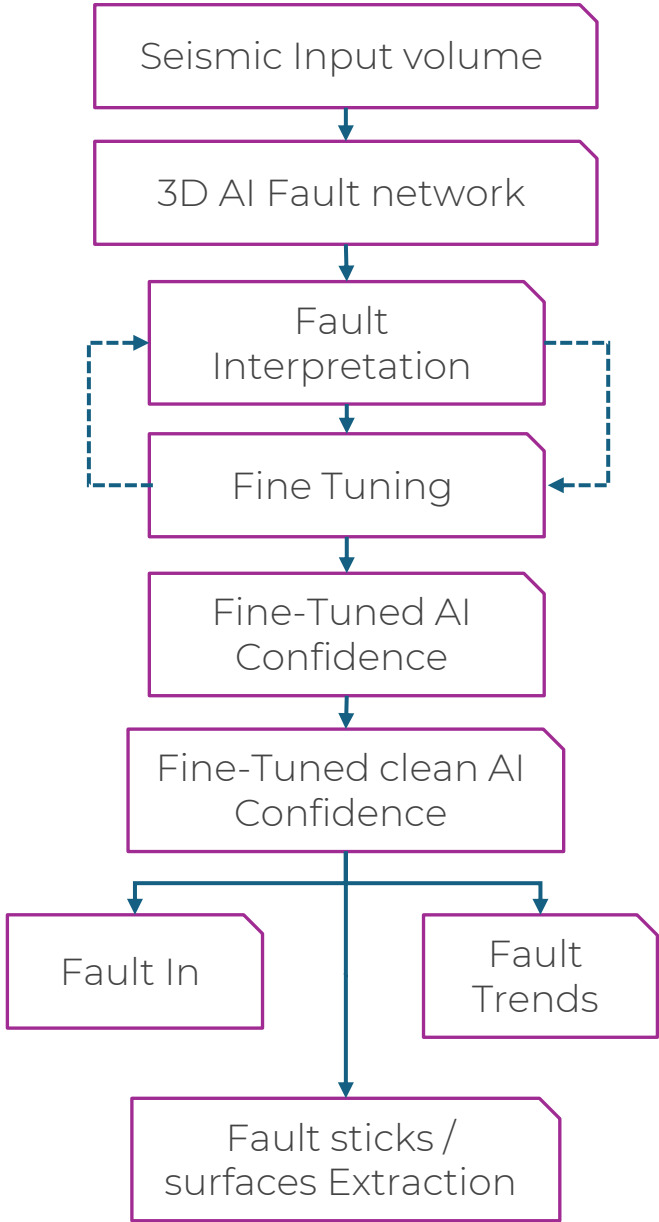


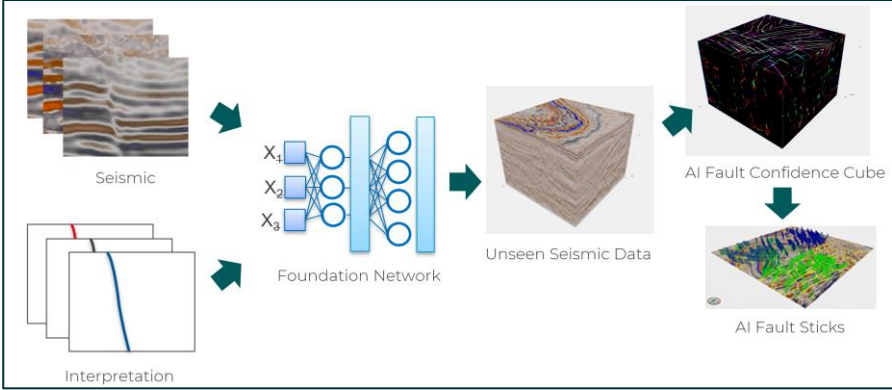
AI Faults Workflow



AI Faults Workflows



Methodologies / algorithms CNNs



Network Name	3D or 2D	Pre-trained	Fine-tuneable	Where to use it?
Birch	3D	Pre-trained	Z sticks	Medium and large-scale faults
Larch	3D	Pre-trained	Z sticks	Faults of all scales
Meranti	3D	Pre-trained	XY sticks	Faults of all scales
Confidence	3D	Pre-trained	No	Mid and large-scale faults
Ash	3D	Pre-trained	No	Small scale, polygonal faults
Acorn	2D	Untrained	XY sticks	Trainable on your own data
AI Network	2D	Pre-trained	XY sticks	For very good data quality

Generated on HighResBD_FT-00-03-a_4_v2 and GT_AI_FT-00-03-a

AI Faults Workflows

Seismic Input volume

3D AI Fault network

Fault Interpretation

Fine Tuning

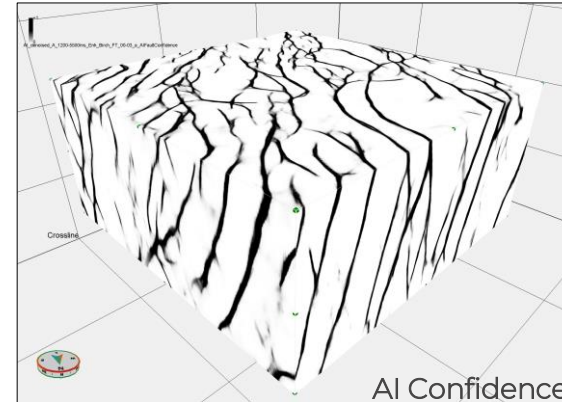
Fine-Tuned AI Confidence

Fine-Tuned clean AI Confidence

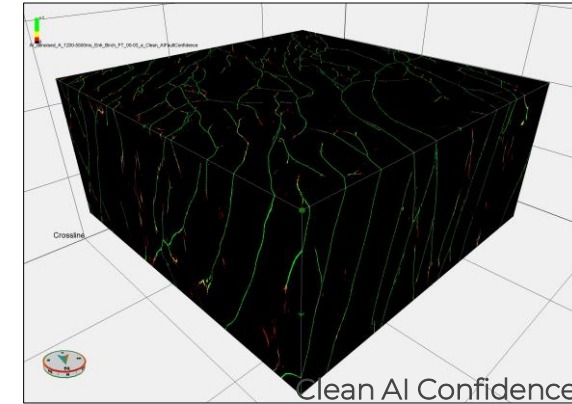
Fault In

Fault Trends

Fault sticks / surfaces Extraction

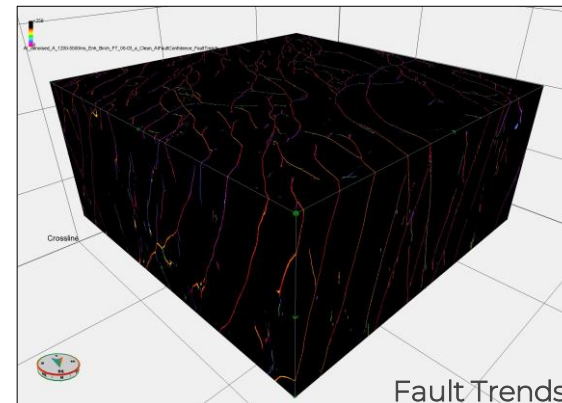


The AI Fault Confidence volume contains the raw numerical values based on the AI confidence (0-100%) that a fault is present in the input seismic data.

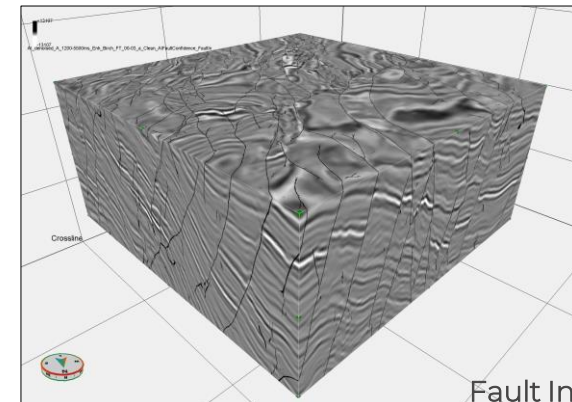


The cleaning process utilises a gaussian smoothing operator followed by Fault Detect. The Fault Detect uses a ridge detection operator and is applied to segment the lineations that are likely to represent fault edges.

The value (10-100%) of the Fault Detect lineation at any point represents the likelihood of the fault edge present within the data.



The Fault Trends process estimates the lateral orientation (strike) of the edge at each point along the detected edge lineation. The result gives us an indication of the families of faults present based on their orientation.



The Fault In process uses the detected faults, assigns them a value at the top end of the dynamic range of the data and embeds them into the Data Conditioned volume. This allows for the interpreter to clearly look at the relationship between the detected faults and the reflectivity data.

Fault Interpretation workflow

